

Marketing Campaign Performance Analysis Report

Project Overview

Project Title

Marketing Campaign Performance Analysis Dashboard

Objective

The objective of this project was to analyze marketing campaign performance data to identify:

- High-performing marketing channels
- Profitable customer segments
- ROI trends over time
- Campaign effectiveness
- Location-based performance
- Business optimization opportunities

The project was designed from the perspective of a Marketing Data Analyst to help marketing agency founders and business decision-makers optimize advertising strategies using data-driven insights.

Tools & Technologies Used

Tool	Purpose
Excel	Data cleaning and preprocessing
SQL (MySQL)	Data querying and business analysis
Python (Pandas, Matplotlib, Seaborn)	Exploratory data analysis and visualization
Power BI	Interactive dashboard creation and business reporting

Dataset Overview

The dataset contains marketing campaign information including:

- Campaign ID
- Company Name
- Campaign Type
- Target Audience

- Marketing Channel
- Conversion Rate
- ROI
- Clicks
- Impressions
- Engagement Score
- Customer Segment
- Acquisition Cost
- Profit Index
- Location
- Date

The dataset was used to evaluate campaign performance across multiple dimensions.

Data Preparation Process

1. Excel Data Cleaning

The raw dataset was cleaned in Excel before analysis.

Cleaning Activities Performed

- Removed duplicate values
 - Checked missing values
 - Verified column consistency
 - Corrected formatting issues
 - Standardized column naming conventions
 - Verified numerical fields
 - Prepared CSV file for SQL import
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2. SQL Analysis

The cleaned dataset was imported into MySQL for business analysis.

SQL Analysis Objectives

- Calculate average ROI
- Identify top-performing channels
- Analyze customer segments
- Evaluate campaign profitability
- Compare locations
- Analyze monthly trends

Sample Business Queries

- Average ROI by channel
 - Most profitable customer segments
 - Top profitable campaigns
 - Monthly ROI trend
 - Location-wise performance analysis
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3. Python Analysis

Python was used for deeper analytical insights.

Libraries Used

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pandas  
matplotlib  
seaborn
```

Analytical Tasks Performed

- Correlation analysis
- Trend analysis
- Customer segmentation analysis
- ROI comparison
- Visualization creation
- Profitability analysis

Visualizations Created

- ROI comparison charts
 - Monthly trend analysis
 - Segment analysis charts
 - Correlation heatmap
 - Channel performance analysis
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Dashboard Overview

The final dashboard was created in Power BI to provide interactive business insights.

Dashboard Components

KPI Cards

The dashboard contains the following KPI metrics:

KPI	Value
Average ROI	5.00
Average Conversion Rate	8.01%
Average Profit Index	50.03K
Total Impressions	1.1 Billion
Total Clicks	110 Million

Key Insights

1. Marketing Channel Performance Analysis

Observation

The dashboard shows that:

- Facebook generated the highest average ROI
- Website and Google Ads also performed strongly
- Instagram produced the lowest ROI among major channels

Business Insight

The organization should prioritize budget allocation toward:

- Facebook campaigns
- Website marketing
- Google Ads optimization

Lower-performing channels such as Instagram may require:

- Better audience targeting
- Creative optimization
- Budget reduction

Recommendation

Increase investment in high-performing channels while re-evaluating low-performing campaign strategies.

2. Customer Segment Analysis

Observation

The Foodies and Fashionistas customer segments generated the highest average profit index.

Health & Wellness generated comparatively lower profitability.

Business Insight

Different customer segments respond differently to marketing campaigns.

Recommendation

Focus future campaigns on:

- Food-related audiences
- Fashion-focused users
- High-engagement customer groups

Lower-performing segments should be analyzed further to improve campaign relevance.

3. Monthly ROI Trend Analysis

Observation

The monthly ROI trend remains relatively stable throughout the year with small fluctuations.

ROI peaks were observed around:

- May
- September

Business Insight

The company maintains relatively consistent campaign performance across the year.

However, certain periods generate stronger returns.

Recommendation

- Increase campaign spending during high-performing months
 - Investigate seasonal behavior patterns
 - Launch premium campaigns during strong ROI periods
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4. Location-Based Performance Analysis

Observation

Location-based analysis shows relatively balanced profitability across cities including:

- Chicago
- Houston
- Los Angeles
- Miami
- New York

Business Insight

Marketing campaigns perform consistently across locations.

No single location dramatically underperforms.

Recommendation

- Continue multi-location campaign strategy
 - Introduce geo-targeted campaigns for deeper optimization
 - Test localized advertisements for specific cities
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5. Company Performance Analysis

Observation

The company comparison table indicates similar ROI values across organizations.

TechCorp and Alpha Innovations demonstrate slightly stronger profit performance.

Business Insight

Most companies maintain comparable campaign effectiveness.

Small ROI differences can still significantly affect long-term profitability.

Recommendation

- Benchmark top-performing companies
- Analyze campaign creatives used by stronger performers
- Optimize underperforming campaigns using best-performing patterns

Business Problems Solved

This project helps solve several business problems:

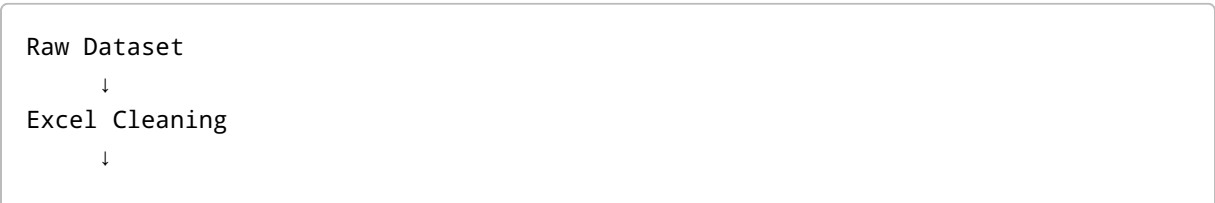
Business Problem	Solution Provided
Poor marketing budget allocation	ROI-based channel analysis
Unclear audience targeting	Customer segment analysis
Campaign inefficiency	Profitability evaluation
Lack of strategic insights	Dashboard-based decision support
Difficulty tracking performance	Interactive KPI reporting

Analytical Approach

The project followed a structured analytical workflow:

1. Data Collection
2. Data Cleaning
3. SQL Analysis
4. Exploratory Data Analysis
5. Visualization
6. Dashboard Development
7. Business Insight Generation
8. Strategic Recommendations

Project Workflow



SQL Analysis



Python Visualization



Power BI Dashboard



Business Insights & Recommendations

Dashboard Features

Interactive Slicers

The dashboard includes dynamic filters for:

- Company
- Customer Segment
- Location

These slicers allow users to interactively explore data.

Advanced Analytical Features

The project includes:

- KPI tracking
 - Trend analysis
 - Segment analysis
 - Channel comparison
 - Profitability analysis
 - Interactive filtering
 - Executive-level visualization
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Strategic Recommendations

Marketing Strategy Recommendations

1. Increase Focus on High-ROI Channels

The organization should:

- Increase budget allocation to Facebook and Website channels
 - Optimize Google Ads campaigns further
 - Reduce unnecessary spending on lower-performing channels
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2. Customer Segment Optimization

Future campaigns should focus on:

- Foodies
- Fashionistas
- Outdoor audience groups

These segments demonstrate stronger profitability potential.

3. Seasonal Campaign Planning

The organization should:

- Identify seasonal campaign opportunities
 - Scale campaigns during high-performing periods
 - Use historical ROI trends for forecasting
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4. Performance Monitoring

The dashboard should be monitored regularly to:

- Detect performance decline
 - Identify emerging opportunities
 - Improve campaign effectiveness
 - Optimize marketing expenditure
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Conclusion

This project successfully analyzed marketing campaign performance using an end-to-end data analytics workflow.

The analysis identified:

- High-performing marketing channels
- Profitable customer segments
- Monthly ROI patterns
- Company-wise profitability insights
- Business optimization opportunities

The Power BI dashboard provides an interactive platform for decision-makers to monitor campaign performance and make strategic marketing decisions.

This project demonstrates practical skills in:

- Data Cleaning
- SQL Analysis
- Python Analytics
- Data Visualization
- Business Intelligence
- Dashboard Development
- Business Insight Generation

Final Outcome

The project delivers:

- ✓ Interactive Power BI Dashboard
 - ✓ Marketing ROI Analysis
 - ✓ Customer Segmentation Insights
 - ✓ Business Decision Support
 - ✓ End-to-End Data Analytics Workflow
 - ✓ Portfolio-Ready Data Analyst Project
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Author

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Data Analyst Project

Marketing Campaign Performance Analysis